The role of troponin C in modulating the Ca^{2+} sensitivity of mammalian skinned cardiac and skeletal muscle fibres

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1. We investigated the effects of acidosis, inorganic phosphate (P_i) and caffeine on the Ca^{2+} affinity of isolated fast-twitch skeletal and cardiac troponin C (TnC), labelled with fluorescent probes to report Ca^{2+} binding to the regulatory sites. We also measured the effects of these interventions on the maximum force development and the Ca^{2+} sensitivity of skinned fibres from fast-twitch skeletal muscle and cardiac muscle, as has been done previously. The two types of experiment were carried out under similar solution conditions, so that we could assess the contribution of any direct actions on TnC to the modulation of Ca^{2+} sensitivity in the skinned muscle fibres.

2. In skinned fibres, acidosis (decreasing pH from 7·0 to 6·2) and P_i (20 mM) suppressed maximum force to the same extent within a given muscle type, but had greater effects on cardiac fibres compared with skeletal fibres. Caffeine (20 mM) depressed maximum force equally in cardiac and skeletal muscle. Thus, the fall of force induced by acidosis or P_i may involve a different mechanism from that induced by caffeine.

3. Skinned skeletal fibres were more Ca^{2+} sensitive than cardiac fibres by 0·29 pCa units ($pCa = -\log_{10}[Ca^{2+}]$). Isolated skeletal TnC also had a greater Ca^{2+} affinity than cardiac TnC, by 0·20 pCa units. These results suggest that the Ca^{2+} sensitivity of skinned fibres is at least partly determined by the type of TnC present.

4. Acidosis reduced the Ca^{2+} sensitivity of force in skinned fibres profoundly and had a 2-fold greater effect in cardiac muscle than skeletal muscle (falls in pCa for 50% activation, pCa_{50}, were 1·09 and 0·55, respectively). Acidosis also reduced the Ca^{2+} affinity of TnC, again having double the effect on the pCa_{50} for cardiac TnC (0·58) as on that for skeletal TnC (0·28). The greater effect of acidosis on cardiac skinned fibres, compared with skeletal, may be partly explained, therefore, by the type of TnC present, and one-half of the effect on fibres may be attributed to the direct effect of H^+ on TnC.

5. P_i reduced the Ca^{2+} sensitivity of force in skeletal and cardiac skinned fibres by 0·30 and 0·19 pCa units, respectively. However, the Ca^{2+} affinity of isolated cardiac and skeletal TnC was unaffected by P_i, indicating that the decrease in muscle Ca^{2+} sensitivity is not mediated by a direct action of P_i on TnC.

6. Caffeine increased the Ca^{2+} sensitivity of cardiac skinned fibres by 0·31 pCa units, which was 3 times greater than for the skeletal fibres (0·09 pCa units). Caffeine had no effect on Ca^{2+} binding to isolated TnC, so its Ca^{2+} sensitizing action on skinned fibres is unlikely to be due to a direct effect on TnC.

7. We conclude that the different Ca^{2+} and pH sensitivities of cardiac and skeletal fibres are partly accounted for by the properties of the TnC present. However, the actions of caffeine and P_i on skinned fibres are not explained by any direct actions on TnC. Some of the P_i-induced decrease in Ca^{2+} sensitivity probably results from a reduced Ca^{2+} affinity of TnC in the thin filament as a result of a decrease in the proportion of attached, force-generating cross-bridges. However, an additional, kinetic mechanism is that there may be a functional antagonism between Ca^{2+} and P_i at the force-generating step in the cross-bridge cycle, and a simple cross-bridge model is presented to illustrate this.

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The primary regulator of contractile force in striated muscle is the concentration of Ca^{2+} in the cytosol during the twitch. However, force production is also dependent upon the sensitivity of the myofibrils to Ca^{2+}, and this can alter under a variety of physiological and pathological conditions. For example, the fall of force during ischaemia in cardiac muscle or fatigue in skeletal muscle may be due, at least in part, to a decrease in myofibrillar responsiveness to Ca^{2+}, brought about by an intracellular acidosis or an increase in the concentration of inorganic phosphate (P_i); both factors act directly on the myofibrils to reduce both their maximum force production and Ca^{2+} sensitivity (e.g. Fabiato & Fabiato, 1978; Cooke & Pate, 1985; Kentish, 1986; Godt & Nosek, 1989). On the other hand, changes in Ca^{2+} sensitivity may prove to be useful therapeutically, in that a novel class of positively inotropic drugs, the 'Ca^{2+} sensitizer drugs' (exemplified by caffeine), can increase myocardial force production by raising myofibrillar Ca^{2+} sensitivity (e.g. Lamont & Miller, 1992).

How agents such as protons, P_i or caffeine affect the Ca^{2+} sensitivity of myofibrils is not clear. One possibility is that they alter the Ca^{2+} affinity of the regulatory myofibrillar protein, troponin C (TnC). TnC has four Ca^{2+} binding sites (I–IV), two of which (sites III and IV) bind both Ca^{2+} and Mg^{2+}, while sites I and II are relatively specific for Ca^{2+} (Potter & Gergely, 1975). In skeletal TnC, the two Ca^{2+}-specific sites have been judged from kinetic and steady-state data to be the ones involved in the regulation of myofibrillar activity (Johnson, Charlton & Potter, 1979). In cardiac TnC, the Ca^{2+} affinity of site I is much reduced owing to amino acid substitution in the Ca^{2+} binding site, leaving only site II with the appropriate Ca^{2+} affinity for myofibrillar regulation (Holroyde, Robertson, Johnson, Solaro & Potter, 1980; Johnson, Collins, Robertson & Potter, 1980).

The effects of pH on Ca^{2+} binding to TnC have been studied previously with isolated troponin (Stull & Buss, 1978; Ogawa, 1985) and intact myofibrils (Blanchard & Solaro, 1984; Blanchard, Pan & Solaro, 1984), but in these studies it was possible to measure only the total binding of Ca^{2+}, which probably included binding of Ca^{2+} to the non-regulatory sites on TnC (Ca^{2+}–Mg^{2+} sites and very low-affinity Ca^{2+}-specific sites), and to sites on other proteins when the measurements were done with myofibrils. Thus it was difficult to assign the observed Ca^{2+} binding, or any alterations in binding due to pH changes, unequivocally to effects on the Ca^{2+}-specific regulatory site(s) on TnC. The direct effects of P_i or caffeine on Ca^{2+} binding to the regulatory sites on TnC have not been investigated.

In the present study, we used the technique of Johnson and co-workers (Johnson, Collins & Potter, 1978; Johnson *et al.* 1980) to measure Ca^{2+} binding specifically to the regulatory Ca^{2+}-specific site(s) on cardiac and fast skeletal TnC. The fluorescent labels IAANS (2-(4′-(iodoacetamido)analino)naphthalene-6-sulphonic acid) and DANZ (dansylaziridine), when bound to TnC near the Ca^{2+}-specific sites, show an increase of fluorescence when Ca^{2+} binds to these sites, and thereby give a measure of Ca^{2+} binding. These probes have been used to study Ca^{2+} binding to isolated TnC and to TnC after reincorporation into myofilaments; the latter procedure has the attraction that TnC is in its native environment and is subject to the normal influences of other myofilament proteins. This is important, since the affinity of the Ca^{2+}-specific sites increases 10-fold on the binding of troponin I to TnC (Potter & Gergely, 1975), decreases 10-fold on the interaction of troponin with actin (Zot, Iida & Potter, 1983) and increases again when myosin binds to the thin filaments (Grabarek, Grabarek, Leavis & Gergely, 1983; Güth & Potter, 1987). However, it is difficult to interpret the results of Ca^{2+}-binding studies using fluorescently labelled TnC reconstituted into myofilaments, because an observed fluorescence change may be due either to the binding of Ca^{2+} to TnC or to protein interactions within the myofibril: for example, the fluorescence increases when rigor cross-bridges are formed even if Ca^{2+} is absent (Güth & Potter, 1987).

To avoid these complexities, we have used isolated, labelled TnC to investigate the direct effects of acidosis, phosphate and caffeine on the Ca^{2+} affinity of the regulatory Ca^{2+}-specific sites. With isolated TnC, the fluorescence change is caused only by the binding of Ca^{2+}. We compared any changes in the Ca^{2+} affinity of cardiac and skeletal TnC with the changes in the Ca^{2+} sensitivity of myofibrillar force development in skinned cardiac and fast skeletal muscles, measured under similar conditions.

Some of these results have been communicated previously (Kentish & Palmer, 1989).

METHODS

Skinned muscle experiments

Apparatus and preparation of skinned fibres. The muscle baths and associated equipment for the skinned muscle experiments have been described in detail previously (Kentish, 1991). In brief, fibres were mounted between carbon fibre arms attached to a servo-motor (300S, Cambridge Technology, Watertown, MA, USA) and to an isometric force transducer (801, SensoNor, Horten, Norway). Solutions in the muscle baths were maintained at $25{\cdot}0 \pm 0{\cdot}1$ °C using a thermoregulator and were stirred by a jet of water-saturated N_2. Sarcomere length (SL) was measured continuously by laser diffraction using a Reticon photodiode array. Muscle force, muscle length and SL signals were recorded on a four-channel chart recorder (2400S, Gould, Hainault, UK).

New Zealand White rabbits (~2·5 kg) were killed by overdose of pentobarbitone (75 mg kg^{-1}). Bundles of psoas fibres were dissected, glycerinated according to the method of Ferenczi, Homsher & Trentham (1984), and kept at −20 °C. Single fibres were dissected from the fibre bundle and were glued with cellulose acetate in acetone to the motor and transducer arms. The fibre was immersed in relaxing solution (see below) for >10 min to wash out the glycerol and stretched until the resting SL was 2·4 μm.

Detergent-skinned ventricular trabeculae were prepared as described previously (Kentish, 1986, 1991). In summary, male Wistar rats (~200 g) were killed by cervical dislocation and the hearts were placed in Tyrode solution. Trabeculae were dissected from the right ventricle and were skinned by 30 min immersion in relaxing solution plus 1% Triton X-100. The muscles were attached to the force transducer and the motor arm using nylon snares. Resting SL was set to 2·1 μm.

Solutions. Control solutions contained (mM): Bes (*N*,*N*-bis(2-hydroxyethyl)-2-aminoethanesulphonic acid), 100; EGTA, 10; total calcium, 0–10·5 (pCa = 8–3·4, where $pCa = -\log_{10}[Ca^{2+}]$); Na_2H_2ATP, 5; $MgCl_2$, 4·7–5·9 ($[Mg^{2+}] = 1·0$ mM); sodium phosphocreatine, 10; potassium propionate, 55; ionic strength, 0·20 M; pH 7·00 at 25 °C. This composition was similar to that used previously (Kentish, 1986, 1991), except that the $[Mg^{2+}]$ was decreased from 3 to 1 mM, and dithiothreitol was omitted to avoid interaction with the fluorescent labels used in the troponin C experiments. For solutions of pH 6·20, the Bes was replaced by 100 mM Mes (2-(*N*-morpholino)ethanesulphonic acid). When P_i (20 mM) was added, the [propionate] was reduced by 30 mM to maintain the ionic strength. Most of the stock solutions were passed through Chelex-100 to remove contaminant Ca^{2+}. The measured purity of EGTA was 96·3% and the contaminant $[Ca^{2+}]$ was 3 μM (measured using fluo-3). Details of solution manufacture and of the calculation of the free ion concentrations have been described previously (Kentish, 1986, 1991). Three types of solution were used, in the following order: relaxing solution (10 mM EGTA, no added Ca^{2+}); preactivating solution (0·15 mM EGTA, 9·85 mM 1,6-diaminohexane-*N*,*N*,*N*′,*N*′-tetracetic acid (HDTA), no added Ca^{2+}), which decreased the Ca^{2+} buffering of the solution inside the muscle and thereby accelerated the subsequent activation of the muscle; and activating solution (10 mM EGTA, 2·5–10·5 mM added Ca^{2+}). The bathing solutions were stored frozen at −20 °C for up to 2 months.

Chemicals were from BDH (Poole, UK), Sigma (Poole, UK) or Fluka (Glossop, UK).

Protocol. A few contractures were elicited using control maximum activating solution (pH 7·0, pCa 3·7) to determine the stability of the fibres. Preparations were rejected (i) if force declined during a contracture or by >20% during the experiment, or (ii) if the resting SL changed after a contracture or became non-homogeneous along the fibre length. The effects of the three interventions (pH 6·2, P_i and caffeine) were examined on each fibre using three series of solution changes. In the first and third series, control and test activating solutions were applied alternately, in order of increasing $[Ca^{2+}]$, with relaxation of the fibre between each activation. Maximum force was measured after every four activations. The procedure for the second series was the same except that the control solutions were omitted: it was found in practice that the control Ca^{2+} sensitivity of the skinned fibres did not alter significantly during an experiment, so control data from the first and third series were averaged and used as the controls for the second series. The order of the different test conditions was rotated between experiments.

Forces were expressed relative to the maximum force in control activating solution (pCa 3·7). The frequent application of this solution to the muscle allowed correction for the small decline (assumed to be linear) in maximum force during an experiment. The relationships between relative force and pCa in each muscle were fitted by non-linear least-squares regression to a Hill equation: relative force = maximum force × $[Ca^{2+}]^{n_H}/K^{n_H} + [Ca^{2+}]^{n_H}$), where n_H is the Hill coefficient and K is a dissociation constant. The pCa value at 50% relative force (pCa_{50}) was then calculated ($= -\log K$). The mean values for pCa_{50} and n_H were used to generate the sigmoid curves shown in the figures.

Measurement of calcium binding to isolated troponin C

Ca^{2+} binding was assessed from the increase in fluorescence of labelled TnC as $CaCl_2$ was added sequentially to solutions containing EGTA as the Ca^{2+} buffer.

Preparation of labelled troponin C. Skeletal TnC was prepared from the back muscles of New Zealand White rabbits (killed by pentobarbitone overdose) according to the method of El-Saleh & Solaro (1988) and was labelled with DANZ by the method of Johnson *et al.* (1978). Cardiac troponin was isolated from cow hearts (obtained from the slaughterhouse) according to the method of Tsukui & Ebashi (1973), except that LiCl was added only once to the regulated actin with a single centrifugation at 7000 *g*. Cardiac TnC was obtained through fractionation of troponin as performed by Greaser & Gergely (1971), except that NaCl was substituted for KCl in the salt gradient and diethylaminoethyl (DEAE) cellulose (Whatman DE-52) for Sephadex in the column. Cardiac TnC was labelled with IAANS by the method of Johnson *et al.* (1980). The labelled proteins were stored concentrated in buffer solution (67 mM KCl + 100 mM Bes, pH 7·0; Ca^{2+} free) at −20 °C.

Solutions. To facilitate comparison with earlier studies, which used simple solutions of ionic strength of approximately 0·1 M, we initially used a solution for the TnC titration that contained (mM): Bes, 100; KCl, 67; EGTA, 2; brought to pH 7·0 with Ca^{2+}-free KOH (ionic strength = 0·1 M). In the main set of experiments the aim was to compare directly the results from isolated TnC and from skinned fibres. Accordingly, the titration solutions were the same as the relaxing solutions used for the skinned fibres (see above), except that the [EGTA] was lowered from 10 to 2 mM, in order to limit the fall in pH as Ca^{2+} was added to EGTA, and the [propionate] was increased to maintain the ionic strength at 0·2 M. Solutions were prepared from the same stocks as those used for skinned fibre work.

Protocol. Fluorescence was measured in a Perkin Elmer LS-5 fluorimeter using the optimal excitation/emission wavelengths of 340/520 nm for DANZ-labelled skeletal TnC (TnC_{DANZ}) and 328/450 nm for IAANS-labelled cardiac TnC (TnC_{IAANS}). The titrations were carried out in a quartz cuvette with continuous stirring. TnC was added to the appropriate solution to a final concentration of 3 μM. (The fluorescence of the solution alone was negligible, except for the solutions with 20 mM caffeine, which at the TnC_{DANZ} wavelength settings had a fluorescence that accounted for ~5% of that recorded after the addition of TnC). Successive aliquots of 100 mM or 1 M $CaCl_2$ (BDH Ltd) were then added to the cuvette with a 1 μl Hamilton syringe. Fluorescence readings were time averaged over 4 s.

The addition of $CaCl_2$ to the EGTA-containing solutions caused a small fall in solution pH. In separate experiments, we measured the exact size of the pH change after each addition, using a Corning 250 pH meter and larger volumes of solutions (without TnC). The measured decreases in pH were taken into account in the calculations of $[Ca^{2+}]$, but were too small (< 0·08 pH units) to affect the Ca^{2+} affinity of the TnC itself. The dilution of the TnC solution caused by addition of $CaCl_2$ was too small (maximum ~3% at pCa 2) to significantly affect the fluorescence readings. Calculated $[Mg^{2+}]$ varied by < 0·1 mM over the entire $[Ca^{2+}]$ range used.

During the titrations, as $[Ca^{2+}]$ increased over the range where Ca^{2+} binds to the Ca^{2+}-specific site (approximately pCa 6–3·5), the fluorescence of the DANZ and IAANS labels increased sigmoidally (e.g. Fig. 3). The absolute fluorescence readings in each solution were converted to relative fluorescence using the formula:

$$\text{Relative fluorescence} = (F_x - F_{min})/(F_{max} - F_{min}),$$

where F_x is the fluorescence at pCa x, F_{min} is the minimum fluorescence in the pCa range 6–8, and F_{max} is the fluorescence at pCa 3·8. The relative fluorescence– pCa curves over the pCa range 6·5–3·8 were then fitted to the Hill equation using non-linear regression. For TnC_{IAANS}, it was impossible to define unambiguously the maximum and minimum fluorescence relating to Ca^{2+} binding in this pCa range, because the minimum and maximum fluorescences as defined above may have included components from overlapping fluorescence changes caused by Ca^{2+} binding to other sites (Fig. 3). To allow for this, during curve fitting the minimum and maximum fluorescences were allowed to vary. (However, the minima and maxima derived from the best fits were never more than 10% different from values as defined above.) On each day, the titrations were repeated once or twice and the average values of n_H and K calculated from the sigmoid fits were used as single data points in the statistical analysis.

Statistical analysis

Unless stated otherwise, values are given as means ± S.E.M. of n experiments. Student's paired or unpaired t tests were applied where appropriate and a significant difference was taken as $P < 0·05$. Using the Kolmogorov–Smirnov test, we ascertained that n_H and pCa_{50} were normally distributed, since this is a prerequisite for using the t test. The pCa_{50} values were a better fit to the normal distribution than were the corresponding $[Ca^{2+}]$ values, indicating that statistics should be performed using the pCa_{50} values rather than the $[Ca^{2+}]$ values.

RESULTS

Force production of myofibrils in skinned fibres

Control solutions

During full Ca^{2+} activation in control solutions, rabbit psoas fibres (mean diameter 65 ± 7 μm, $n = 5$) generated a stress of 84·4 ± 15·6 mN mm^{-2} and rat trabeculae (mean diameter 134 ± 20 μm, $n = 5$) produced 25·0 ± 7·2 mN mm^{-2}. These values were similar to those found in previous studies (e.g. Brandt, Diamond & Schachat, 1984; Kentish, 1986). As shown in Fig. 1, in all fibres the relationship between relative force and pCa was sigmoidal and was fitted well by the Hill equation. Details of the force–pCa relationships are given in Table 1. Under control conditions the pCa required for 50% activation (pCa_{50}) was 5·57 in rat trabeculae and 5·86 in rabbit psoas fibres. Thus, fast skeletal fibres were more sensitive than cardiac muscles to Ca^{2+} by 0·29 pCa units. The slope of the force–pCa relationship was slightly steeper in cardiac muscle than in skeletal muscle, as reflected in the higher Hill coefficient (n_H, Table 1) of 2·72 compared with 1·90.

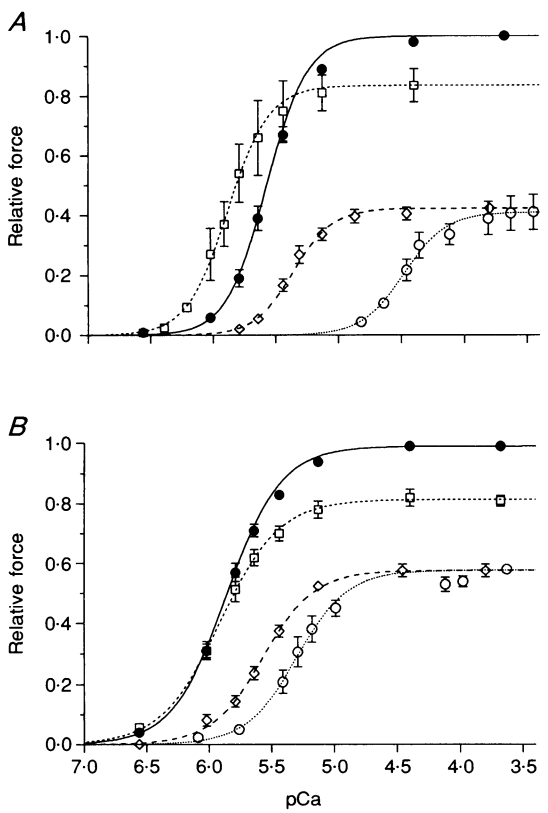


Figure 1. Effects of caffeine, acidosis and P_i on force–pCa relationships in cardiac and fast skeletal skinned fibres
Skinned fibres were prepared from rat ventricle (*A*) and rabbit psoas muscle (*B*). ●, control, pH 7·0; □, 20 mM caffeine; ◇, 20 mM added P_i; ○, pH 6·2. Forces are given relative to the maximum force in control solution. $pCa = -\log_{10}[Ca^{2+}]$. Points show means ± S.E.M. from 5 cardiac and 4 skeletal muscles. The sigmoid curves were generated from the mean values for pCa_{50} and n_H (Table 1) as described in the text.

Table 1. Characteristics of Ca^{2+} activation in skinned fibres from rat ventricular muscle and rabbit psoas muscle under various solution conditions

Solution	n	Maximum force (% control)	pCa_{50}	n_H
Cardiac				
Control	5	100	5·57 ± 0·03	2·72 ± 0·17
pH 6·2	5	41·0 ± 5·8*	4·48 ± 0·01*	2·55 ± 0·14
P_i	5	42·6 ± 1·6*	5·38 ± 0·03*	2·84 ± 0·04
Caffeine	5	84·4 ± 5·6*	5·88 ± 0·05*	2·64 ± 0·15
Fast skeletal				
Control	5	100	5·86 ± 0·06*	1·90 ± 0·10
pH 6·2	4	57·7 ± 1·8*	5·30 ± 0·06*	2·13 ± 0·14
P_i	5	57·6 ± 2·1*	5·57 ± 0·04*	2·10 ± 0·13
Caffeine	4	81·4 ± 2·4*	5·94 ± 0·04*	1·83 ± 0·01

The force–pCa relationship is described by the pCa_{50} (pCa required for 50% activation) and n_H (Hill coefficient, reflecting the slope of the relationship). Control solutions were pH 7·0 with no added P_i or caffeine. The pH was reduced to 6·2, or P_i (20 mM) or caffeine (20 mM) was added. Means ± S.E.M. of n experiments. *Significantly different from control (paired t test, $P < 0·05$).

Acidosis

Decreasing the pH from 7·0 to 6·2 suppressed the maximum Ca^{2+}-activated force in both cardiac and skeletal fibres (Table 1 and Fig. 1). This decrease in force was not due to incomplete Ca^{2+} activation of the myofibrils, resulting from the concomitant decrease in Ca^{2+} sensitivity (see below), since the $[Ca^{2+}]$ had been raised sufficiently high to produce full activation of the myofibrils. Maximum force was suppressed to 41 and 58% of control in cardiac and skeletal fibres, respectively; thus maximum force was more pH sensitive in cardiac muscle. Acidosis, P_i or caffeine had no consistent effects on the resting force observed at pCa > 8 (results not shown).

Acidosis also shifted the force–pCa relationships for both cardiac and skeletal skinned fibres to the right,

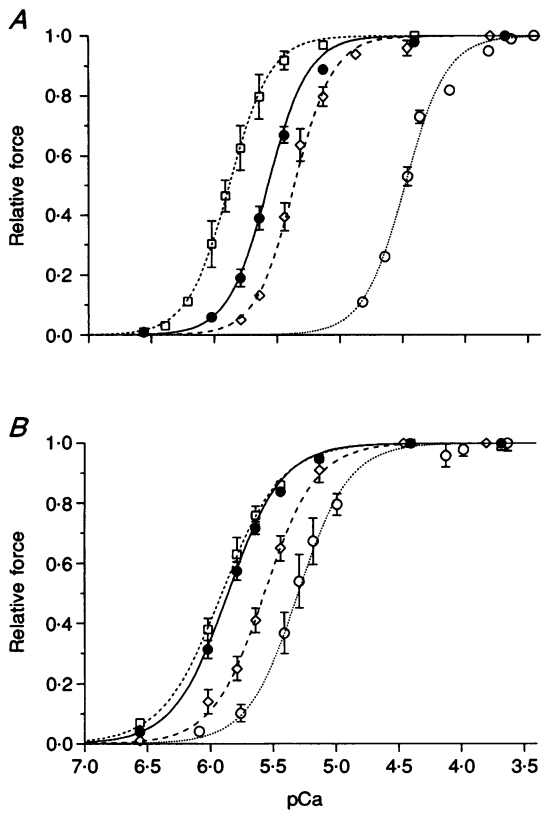


Figure 2. Effects of caffeine, acidosis and P_i on the Ca^{2+} sensitivity of force production in skinned fibres
A, rat ventricle; *B*, rabbit psoas muscle. Same data as in Fig. 1, but with forces in a particular type of solution expressed relative to the maximum force in that solution. ●, control, pH 7·0; □, 20 mM caffeine; ◇, 20 mM added P_i; ○, pH 6·2.

without affecting the slope of the force–[Ca^{2+}] relationship (Table 1). This is shown more clearly in Fig. 2, where the curves for Fig. 1 have been normalized. The shift was large in the cardiac fibres, in which the fall of pCa_{50} was $1{\cdot}09 \pm 0{\cdot}02$ ($n = 5$); this was double that ($0{\cdot}55 \pm 0{\cdot}02$, $n = 4$) with the psoas fibres. These, and other changes in pCa_{50} for skinned fibres and for isolated TnC, are summarized and plotted in Fig. 6.

Inorganic phosphate (P_i)

Maximum force in both muscle types was suppressed by the addition of P_i (20 mM). As was the case with acidosis, the decrease was greater in trabeculae (to 43% of control; Fig. 1) than in psoas muscle (to 58%). Figure 2 shows that the Ca^{2+} sensitivity of the fibres was also reduced by the addition of P_i, but this effect was, in contrast to the effect on maximum force, greater in skeletal than cardiac fibres; the pCa_{50} for trabeculae was reduced by $0{\cdot}19 \pm 0{\cdot}03$ units ($n = 5$), and for skeletal muscle by $0{\cdot}30 \pm 0{\cdot}02$ units ($n = 5$). It is clear from Fig. 2 that in both muscles the P_i-induced fall in pCa_{50} was markedly less than that observed for acidosis, despite the similar effects of P_i and acidosis on maximum force; this suggests there must be some differences in the mechanisms of inhibition of maximum force and of Ca^{2+} sensitivity. The slopes of the force–pCa relationships in psoas fibres and trabeculae were not significantly affected by P_i (Table 1).

Caffeine

Caffeine (20 mM) reduced the maximum force by 16 and 19% in cardiac muscle and skeletal muscle, respectively. Unlike acidosis or P_i, caffeine shifted the force–pCa relationships of cardiac and skeletal muscles to the left, indicating an increase in myofibrillar Ca^{2+} sensitivity (Figs 1 and 2). There was no effect on the steepness of each relationship (Table 1). The rise in pCa_{50} induced by caffeine was three times greater for trabeculae ($0{\cdot}31 \pm 0{\cdot}04$, $n = 5$) than for skeletal fibres ($0{\cdot}09 \pm 0{\cdot}01$, $n = 4$). To see whether this difference was due to the longer SL in the psoas fibres, since it has been reported that the Ca^{2+} sensitization by caffeine is less at longer SL (de Beer, Grundemann, Wilhelm, Caljouw, Klepper & Schiereck, 1988), in separate experiments we varied the SL in skeletal fibres over a wide range (1·6–2·8 μm). However, the increase in pCa_{50} induced by caffeine in psoas muscles was found to be independent of SL (results not shown), suggesting that the greater Ca^{2+}-sensitizing action of caffeine in cardiac fibres compared with skeletal fast-twitch fibres was due to a true difference between muscle types.

Calcium binding to isolated troponin C

To determine to what extent the properties of cardiac and skeletal TnC could account for the actions of H^+, P_i and caffeine on the Ca^{2+} sensitivity of the skinned fibres, and for the differential sensitivities of cardiac and skeletal fibres to these agents, we measured the binding of Ca^{2+} to the Ca^{2+}-specific sites of isolated cardiac and skeletal TnC, using solutions as similar as possible to those used for the skinned fibre experiments.

Control solutions at low ionic strength

The first set of experiments used solutions of low ionic strength (0·1 M) to enable comparison of the results with previous work (Johnson *et al.* 1978, 1980). Figure 3 shows the changes in fluorescence of the labelled skeletal TnC (TnC_{DANZ}) and cardiac TnC (TnC_{IAANS}) as Ca^{2+} was added to these solutions. In this figure, and in Figs 4 and 5, the fluorescence changes are expressed relative to that due to Ca^{2+} binding to the Ca^{2+}-specific site, occurring in the pCa range 6·5–3·8 (see below). The *absolute* increases in fluorescence over this pCa range were, relative to the minimum fluorescence, $89 \pm 2\%$ ($n = 7$) for skeletal TnC and $31 \pm 1\%$ ($n = 6$) for cardiac TnC.

With TnC_{DANZ}, the fluorescence increased in a simple sigmoidal fashion as the [Ca^{2+}] was increased over the pCa range 8·5–3·0. The curve had a pCa_{50} (pCa for 50% of the fluorescence increase) of $5{\cdot}75 \pm 0{\cdot}04$ ($n = 9$), with a n_H of $1{\cdot}70 \pm 0{\cdot}14$. This pCa_{50} was close to those found in previous studies using labelled (Johnson *et al.* 1978) and unlabelled TnC (Potter & Gergely, 1975), where it was shown to correspond to Ca^{2+} binding to the two Ca^{2+}-specific, regulatory sites on skeletal TnC.

The titration curve for cardiac TnC_{IAANS} was more complex: as the [Ca^{2+}] was increased from pCa 8·5, the

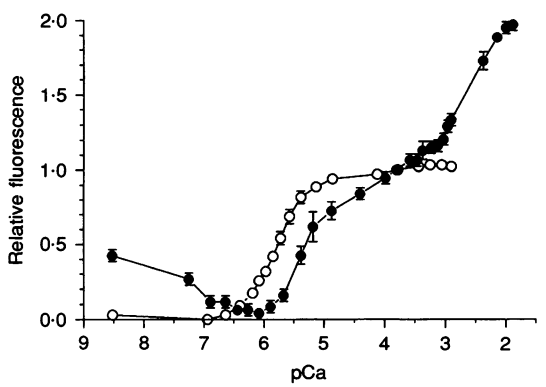


Figure 3. Fluorescence of isolated fluorescently labelled troponin C as a function of solution pCa
Skeletal TnC (○) was prepared from rabbit fast skeletal muscle and was labelled with DANZ; cardiac TnC (●) from bovine ventricular muscle was labelled with IAANS. The change in fluorescence is given relative to the total change occurring between the minimum fluorescence at pCa 6–7 and a maximum, arbitrarily set at pCa 3·8. Solutions were 'low' ionic strength (0·1 M) at pH 7·0, with 3 μM labelled TnC. Points show means ± S.E.M. of 6 cardiac and 4 skeletal experiments.

relative fluorescence at first fell by 40% (mid-point pCa ~7·2), and then rose in two sigmoidal steps, one over pCa 6–3·5 and the other over pCa 3·5–2. Such triphasic behaviour has been found previously by Johnson *et al.* (1980), who demonstrated that, as with skeletal TnC, the increase in fluorescence of cardiac TnC occurring in the pCa range 6·5–3·5 reflects Ca^{2+} binding to the Ca^{2+}-specific sites. As these binding sites are thought to mediate the Ca^{2+} regulation of myofibrils, we concentrated on this pCa range in subsequent experiments. From the sigmoid curves fitted over this pCa range in Fig. 3, the pCa_{50} for the Ca^{2+}-specific sites was found to be 5·30 ± 0·16 ($n = 6$), with a n_H of 1·34 ± 0·27. Thus, the Ca^{2+} affinity of cardiac TnC was considerably less than that of fast skeletal TnC ($pCa_{50} = 5·75$). Although the slopes of the curves were not significantly different, with n_H values of 1·70 ± 0·14 ($n = 9$) for skeletal TnC and 1·34 ± 0·27 ($n = 6$) for cardiac TnC, only the n_H value for skeletal TnC was significantly greater than 1·0 (one-tailed t test, $P < 0·05$).

Control solutions at high ionic strength

In the higher ionic strength (0·2 M) 'physiological' solutions, similar to those used for the skinned muscles, the fluorescence–pCa curves were generally similar to those at low ionic strength. Relative to the minimum fluorescence, the increases in fluorescence over the pCa range 6·5–3·8 were 110 ± 1% ($n = 4$) and 104 ± 2% ($n = 6$) in two batches of TnC_{DANZ}, and 18 ± 1% ($n = 4$) and 31 ± 1% ($n = 9$) in two batches of TnC_{IAANS}. These fluorescence changes were generally suppressed a little by acidosis, caffeine or P_i, except that the TnC_{IAANS} fluorescence was potentiated by acidosis (Table 2). The Ca^{2+} affinity of cardiac TnC (pCa_{50} of 5·28–5·36) was not significantly different from that at low ionic strength (5·30). However, the skeletal curve was shifted to higher [Ca^{2+}], with a mean pCa_{50} of ~5·54 compared with 5·75 at low ionic strength. In spite of this, skeletal TnC was still significantly more Ca^{2+} sensitive than cardiac TnC ($P < 0·05$). Although the n_H values appeared to be similar for skeletal and cardiac TnC, in fact n_H for skeletal TnC was significantly greater than 1·0, whereas that for cardiac TnC was not (Table 2). It may be noted that the pCa_{50} values for Ca^{2+} binding to isolated TnC were lower than those for force production of the corresponding skinned fibre, i.e. Ca^{2+} binding to TnC occurred at higher [Ca^{2+}] than was seen with Ca^{2+} activation of the myofibrils.

Acidosis

Reducing the pH from 7·0 to 6·2 caused rightward displacements of the fluorescence–pCa curves of both TnC_{IAANS} and TnC_{DANZ} (Fig. 4 and Table 2). Thus acidosis decreased the Ca^{2+} affinity of both cardiac and skeletal TnC. The decrease of pCa_{50} for cardiac TnC (0·58 ± 0·05, $n = 4$) was twice as great as for skeletal TnC (0·28 ± 0·01, $n = 4$), as was found with the skinned fibre experiments. However, for both cardiac and skeletal isolated TnC, the pH-induced changes in pCa_{50} were only about one-half of those observed in skinned fibres (Fig. 6). Acidosis also significantly reduced the n_H value for skeletal, but not cardiac, isolated TnC (Table 2). Since EGTA is a rather

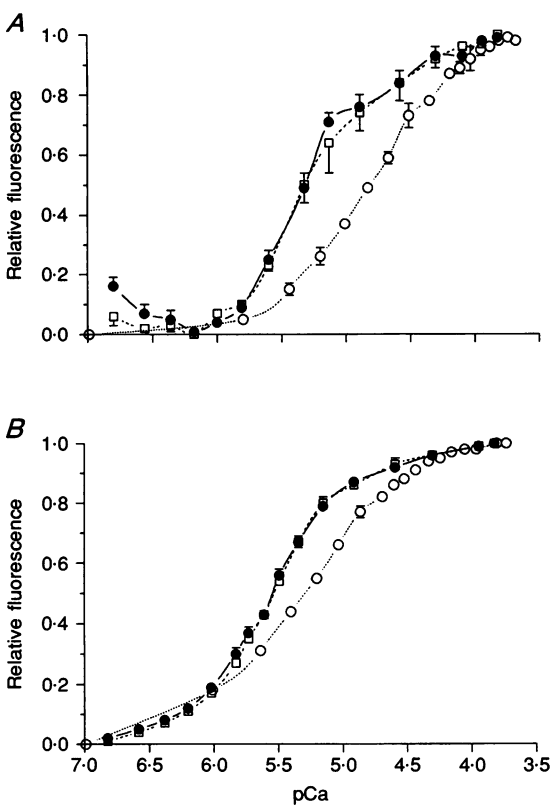


Figure 4. Influence of acidosis or caffeine on Ca^{2+} binding to isolated TnC

A, cardiac TnC_{IAANS}; *B*, skeletal TnC_{DANZ}. ●, control, pH 7·0; □, pH 7·0 with 20 mM caffeine; ○, pH 6·2. Solutions were similar to those used for the skinned fibre studies, with 3 μM TnC added. Points show means ± S.E.M. ($n = 4$).

Table 2. Characteristics of Ca^{2+} binding to isolated, labelled troponin C from bovine heart and rabbit skeletal muscle

Solution	n	Maximum fluorescence change (% control)	pCa_{50}	n_H
Cardiac				
Control A	4	100	5·36 ± 0·06	1·45 ± 0·20
pH 6·2	4	115 ± 3	4·78 ± 0·04*	1·17 ± 0·08
Caffeine	4	94 ± 6	5·37 ± 0·07	1·17 ± 0·28
Control B	4	100	5·28 ± 0·16	1·28 ± 0·22
P_i	4	94 ± 3	5·22 ± 0·05	0·90 ± 0·07
Fast skeletal				
Control A	4	100	5·56 ± 0·02	1·36 ± 0·06†
pH 6·2	4	88 ± 1	5·28 ± 0·01*	1·01 ± 0·02*
Caffeine	4	81 ± 2	5·54 ± 0·01	1·43 ± 0·05†
Control B	4	100	5·51 ± 0·04	1·43 ± 0·10†
P_i	4	94 ± 2	5·48 ± 0·05	1·27 ± 0·09†

The maximum fluorescence change is relative to the maximum change in control solutions within the pCa range 6·5–3·5. Solutions were similar to those for Table 1. Controls A and B refer to control solutions for pH 6·2 or caffeine, and P_i experiments, respectively, Means ± S.E.M. of n experiments. *Significantly different from control (paired t test, $P < 0·05$); †significantly greater than 1·0 (one-tailed t test, $P < 0·05$).

poor Ca^{2+} buffer at low pH, the minimum [Ca^{2+}] tested was at only pCa ~7·0 (where the Ca^{2+} was due to contamination alone). No change in TnC_{DANZ} fluorescence from control was noted at this pCa, but there was an indication that the fluorescence of TnC_{IAANS} seen at pCa > 6·5 in control solution was abolished at pH 6·2 (Fig. 4*A*).

Caffeine

As shown in Figs 4 and 6, the addition of caffeine (20 mM) had no significant effect on the fluorescence–pCa curves of TnC_{IAANS} or TnC_{DANZ} (the only exception to this was a tendency for suppression of TnC_{IAANS} fluorescence at pCa > 6·5). The changes in pCa_{50} were not significant:

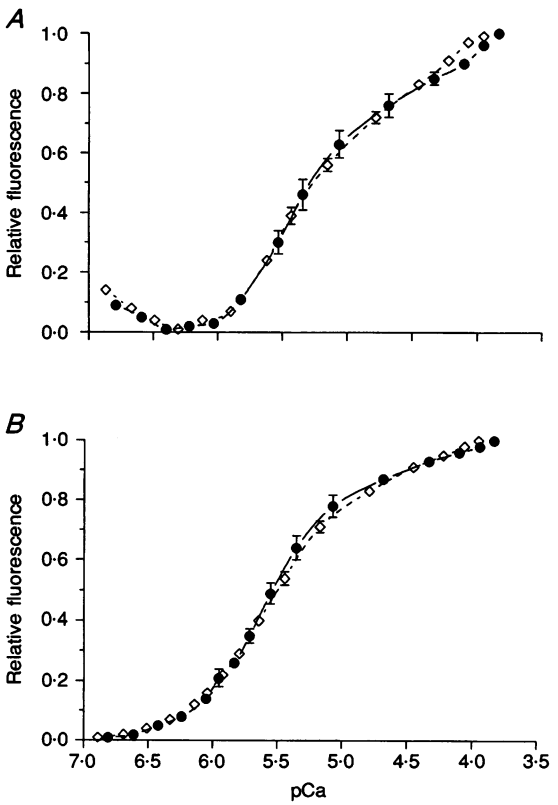


Figure 5. Lack of influence of P_i on Ca^{2+} binding to isolated troponin C

A, cardiac TnC_{IAANS}; *B*, skeletal TnC_{DANZ}. ●, control, pH 7·0; ◇, with 20 mM P_i added. Points show means ± S.E.M. ($n = 4$).

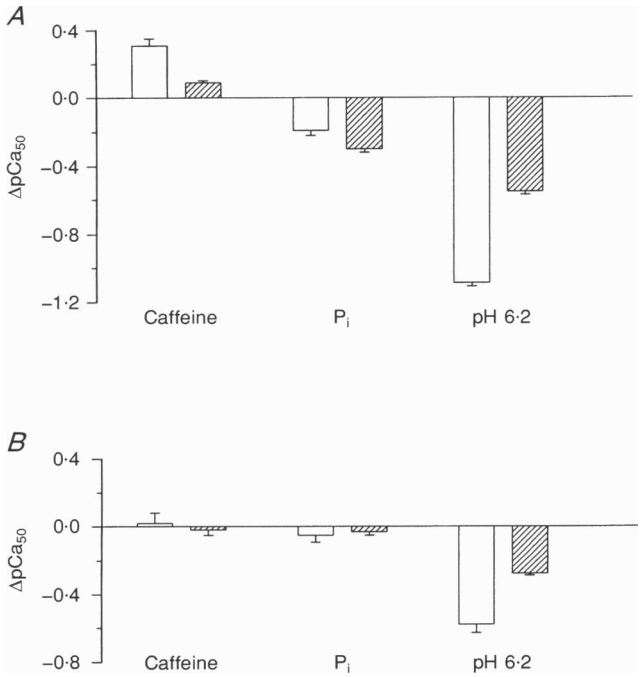


Figure 6. Comparison of the effects of caffeine, P_i and acidosis on the Ca^{2+} sensitivities of skinned fibres and isolated troponin C
Changes in Ca^{2+} sensitivity of skinned fibres (*A*) and Ca^{2+} affinity of TnC (*B*), produced by caffeine (20 mM), P_i (20 mM) or acidic solution (pH 6·2). The bars show the mean changes in pCa_{50} (± S.E.M.) from that in control solution (pH 7·0); □, cardiac; ▨, skeletal.

$0{\cdot}02 \pm 0{\cdot}06$ ($n = 4$) for cardiac TnC and $-0{\cdot}02 \pm 0{\cdot}03$ ($n = 4$) for skeletal TnC. There was no significant change in n_H for either protein (Table 2). This lack of effect of caffeine on binding to the Ca^{2+}-specific site(s) of TnC was in contrast to the Ca^{2+}-sensitizing action of caffeine on skinned cardiac and skeletal fibres, where pCa_{50} increased by 0·31 and 0·09 units, respectively (Fig. 6).

Inorganic phosphate

The effects of P_i addition were studied using different batches of TnC_{DANZ} and TnC_{IAANS} from those used for the above experiments. The results are shown in Fig. 5. P_i (20 mM) had no significant effects on the fluorescence–pCa curves of TnC_{IAANS} or TnC_{DANZ}: the changes in pCa_{50} were $-0{\cdot}05 \pm 0{\cdot}04$ ($n = 4$) for cardiac TnC and $-0{\cdot}03 \pm 0{\cdot}02$ ($n = 4$) for skeletal TnC. Thus the P_i-induced rightward shifts in pCa_{50} (0·19 and 0·30 units) observed in skinned cardiac and skeletal fibres, respectively, were not matched by similar effects of P_i on isolated TnC (Fig. 6).

DISCUSSION

Maximum force production in skinned fibres

Our results showing that P_i, acidosis (pH 6·2) and caffeine decrease the maximum force production of cardiac and fast skeletal myofibrils are broadly in line with those in previous experiments (e.g. Fabiato & Fabiato, 1978; Wendt & Stephenson, 1983; Kentish, 1986; Godt & Nosek, 1989). However, this was the first time that all three interventions were studied together and comparison between their effects on cardiac and fast skeletal muscle (Fig. 1 and Table 1) provides some pertinent information on their likely mechanisms of action. Acidosis had a greater suppressive effect in cardiac than in skeletal skinned fibres (decrease to 41% of control *vs.* 58%). Similarly, the reduction by P_i (20 mM) was greater in cardiac fibres (to 43%) than in skeletal fibres (to 58%), as found by Godt & Nosek (1989). These differential effects between cardiac and fast skeletal muscle were not reproduced by caffeine, which decreased force similarly (to ~83%) in both muscle types. This suggests that caffeine inhibits maximum force by a mechanism different from that for P_i or acidosis. Although it was by chance that the pH 6·2 and 20 mM P_i solutions had the same effect on maximum force in cardiac muscle (Fig. 1), it is of interest that this identity was also seen with the skeletal fibres. This similarity in the actions of these pH and P_i interventions *between* different muscle types is consistent with the idea that acidosis and P_i may have analogous mechanisms of action on maximum force production.

Of the three interventions used here, the mechanism of action of P_i is the best understood. P_i, as a product of ATP hydrolysis, must be released at some step in the cross-bridge cycle of force generation. It appears likely that this step is in fact that at which force is generated: a reversible transition of the weakly bound cross-bridge (AM.ADP.P_i (AM, actomyosin)), which generates little or no force, to the strongly bound cross-bridge (AM.ADP) that generates force (Hibberd, Danzig, Trentham & Goldman, 1985; Pate & Cooke, 1989). An increase in [P_i] will therefore reverse this transition and decrease the proportion of cross-bridges in the force-generating state (correspondingly increasing the proportion of weakly attached and detached states). The greater P_i sensitivity of cardiac fibres compared with

fast skeletal fibres may be a consequence of a lower affinity of M.ADP.P_i (M, myosin) for actin in cardiac muscle (see Kawai, Saeki & Zhao, 1993).

The precise mechanism by which acidosis suppresses maximum force is not known. A proton is, like P_i, a product of ATP hydrolysis and so must be released during the cross-bridge cycle. One possible mechanism for the suppression of force by acidosis is that increased $[H^+]$ reverses the, as yet unidentified, H^+ release step; this would decrease force if this step preceded, or was the same as, the force-generating step of the cross-bridge cycle (discussed in detail by Kentish, 1991). Kentish proposed that H^+ release occurred at a transition between two weakly attached AM.ADP.P_i states preceding the P_i release (force-generating) step, and the results of Seow & Ford (1993) indicate that acidosis indeed acts on a weakly attached cross-bridge state. The resemblance in the proposed mechanisms for the actions of acidosis and P_i is consistent with our finding that the differential action on cardiac and skeletal muscle was the same for acidosis and P_i (Fig. 1).

Even less is known about how caffeine suppresses maximum force. One possibility is that it is due to an increase in myofilament lattice spacing (de Beer *et al.* 1988). It would seem that the depression of force is not related to Ca^{2+} sensitization, since some Ca^{2+} sensitizing agents such as sulmazole enhance maximum force (Lamont & Miller, 1992); in addition, force and Ca^{2+} sensitivity were affected differentially, in that caffeine reduced maximum force similarly in the two muscle types (Fig. 1) but had a 3-fold greater effect on Ca^{2+} sensitivity in cardiac fibres compared with skeletal fibres (Fig. 2). Clearly the mode of action of caffeine on maximum force will require further investigation.

Calcium sensitivity of skinned fibres

In control solutions, skeletal fibres were found to be more Ca^{2+} sensitive (pCa_{50} 5·86) than cardiac fibres (5·57). This difference could be due to a difference either between the two muscle types or between the two species we used (rabbit and rat). However, others have found that, in a given species, the Ca^{2+} sensitivity of fast skeletal fibres is indeed greater than that of cardiac muscles (rat: Wendt & Stephenson, 1983; rabbit: Godt & Nosek, 1989), although Kerrick *et al.* (1980) reported similar Ca^{2+} sensitivities of rabbit skeletal and cardiac fibres. In addition, in other experiments in our laboratory done under slightly different conditions from those reported here, we found that the pCa_{50} of *rat* psoas fibres ($5·79 \pm 0·04$, $n = 11$) was not significantly different from that for rabbit psoas ($5·76 \pm 0·04$, $n = 7$). Thus the higher Ca^{2+} sensitivity of rabbit psoas compared with rat ventricle seen here seems to be a difference between muscle types, rather than between species. This difference may be partly due to the type of TnC present (see next section).

Our results confirm previous findings that acidosis or P_i decreases myofibrillar Ca^{2+} sensitivity (e.g. Fabiato & Fabiato, 1978; Kentish, 1986; Godt & Nosek, 1989; Morano & Rüegg, 1991), whereas caffeine increases it (Wendt & Stephenson, 1983). As in some of these studies, we observed that pH or caffeine had a greater effect on the Ca^{2+} sensitivity of cardiac, compared with skeletal, skinned fibres. However, the smaller effect of P_i (20 mM) on the pCa_{50} of cardiac muscle (0·19) compared with skeletal muscle (0·30) was opposite to the results of Godt & Nosek (1989). The P_i-induced change in pCa_{50} in the cardiac fibres was also smaller than found previously in this laboratory (0·38: Kentish, 1986), but there were solution differences between our two studies (e.g. $[Mg^{2+}]$ was lower in the present work).

Although these effects of pH, P_i and caffeine on Ca^{2+}-regulated force production were largely known from previous work, it was necessary for us to determine the effects of these agents under our conditions so that we could compare the effects with the actions on Ca^{2+} binding to isolated TnC, measured under similar conditions.

The role of troponin C in myofibrillar activation under control conditions

At low ionic strength (0·1 M), the fluorescence increase of labelled TnC over the pCa range 6·5–3·5 has been shown to be due to Ca^{2+} binding to the Ca^{2+}-specific site(s) of skeletal and cardiac TnC (Johnson *et al.* 1978, 1980). Under similar solution conditions, we find a fluorescence increase over this pCa range (Fig. 3) and our pCa_{50} values (5·75 and 5·30 for skeletal and cardiac TnC, respectively) were similar to those obtained by Johnson and co-workers, suggesting we too were studying binding to the Ca^{2+}-specific site(s).

There were notable differences between the fluorescence responses of cardiac and skeletal TnC to changes in $[Ca^{2+}]$ (Fig. 3). One was that, as $[Ca^{2+}]$ was raised above pCa 3·8, the fluorescence increased with cardiac TnC (TnC_{IAANS}) but not with skeletal TnC (TnC_{DANZ}). This increase probably reflects Ca^{2+} binding to the mutated cardiac site I (Johnson *et al.* 1980). This binding is of no physiological importance, since intracellular $[Ca^{2+}]$ never reaches these levels in cardiac muscle. Another difference was the change in TnC_{IAANS} fluorescence that occurred at pCa ~7·2. Since Ca^{2+} binds to the Ca^{2+}–Mg^{2+} sites in cardiac TnC with an association constant (K_a) of 2×10^7 M^{-1} (Holroyde *et al.* 1980), the fluorescence decrease upon Ca^{2+} addition was probably related to Ca^{2+} binding to these sites. Johnson *et al.* (1980) observed a similar decrease in fluorescence of TnC_{IAANS} as Ca^{2+} bound to the Ca^{2+}–Mg^{2+} sites in the absence of Mg^{2+}; however, the drop in fluorescence also occurred when Mg^{2+} bound in the absence of Ca^{2+} (dissociation constant (K_d) ~0·3 mM). Since our solutions contained 1 mM Mg^{2+}, these sites should always have Mg^{2+} or Ca^{2+} bound. Our results would suggest that the binding of Ca^{2+} and Mg^{2+} to the Ca^{2+}–Mg^{2+} sites is not

equivalent, since the replacement of bound Mg^{2+} by Ca^{2+} is the likely cause of the fluorescence decrease we observed at pCa ~7·2 (but cf. Johnson *et al.* 1980).

In the 'physiological' solutions similar to those used for the skinned muscle experiments, the fluorescence change for the Ca^{2+}-specific site of TnC_{DANZ} had a higher Ca^{2+} sensitivity than that for TnC_{IAANS}, since its pCa_{50} was 5·56 compared with the cardiac 5·36 (Fig. 4). Our interpretation is that the Ca^{2+} affinity of the Ca^{2+}-specific site is higher in fast skeletal TnC than in cardiac TnC (but see below). If this is the case, then the 0·20 pCa unit difference could help to explain why fast-twitch skeletal fibres were 0·29 pCa units more Ca^{2+}-sensitive than cardiac fibres (pCa_{50} of 5·86 and 5·57, respectively). This supports the results obtained through TnC substitution in skinned fibres, which suggested that Ca^{2+} sensitivity may be determined by the moiety of TnC present (Moss, Lauer, Giulian & Greaser, 1986). However, the lower Ca^{2+} sensitivity of cardiac muscle could also be partly explained by other factors, such as a lower active SL in cardiac muscle (resting SL was 2·1 μm compared with 2·4 μm in skeletal muscle).

On the other hand, the difference in the pCa_{50} values of skeletal and cardiac TnC could be explained if the fluorescence change in TnC_{DANZ} does not directly reflect *overall* Ca^{2+} binding to both sites, but occurs in full in each TnC_{DANZ} molecule with Ca^{2+} binding to *either* site; in this case the fluorescence could be up to 0·38 pCa units more Ca^{2+} sensitive than the overall Ca^{2+} binding (see Zot, Güth & Potter, 1986; Güth & Potter, 1987). In contrast, cardiac TnC, with only one regulatory site, would show the same values for the pCa_{50} of fluorescence and Ca^{2+} binding (Zot *et al.* 1986). This factor could thus account for the apparent difference between skeletal and cardiac TnC. In support of this view, Zot *et al.* (1986) and Güth & Potter (1987) found that the fluorescence in TnC_{DANZ}-reconstituted skeletal fibres was more Ca^{2+} sensitive than force, and suggested this was evidence that fluorescence increased on Ca^{2+} binding to either site, whereas force needed Ca^{2+} binding to *both* sites. However, this interpretation has now been questioned, and it has been suggested that the different Ca^{2+} sensitivities of fluorescence and force in skeletal fibres may be due to weakly bound cross-bridges, promoting fluorescence change but not contributing to force development (e.g. Morano & Rüegg, 1991).

Another disparity between cardiac and skeletal TnC was that, at both low and physiological ionic strengths, the Hill coefficient (n_H) was significantly greater than 1 for TnC_{DANZ} but not for TnC_{IAANS} (e.g. Table 2). Since n_H usually reflects the minimum number of interacting sites, this suggests there is co-operativity in Ca^{2+} binding to the two Ca^{2+}-specific sites in the skeletal TnC molecule. In fact, Grabarek *et al.* (1983) showed that n_H for fluorescence should be $> 1·2$ before it can be concluded that Ca^{2+} binding is co-operative, but our value of 1·36 for skeletal TnC_{DANZ} is indeed significantly above 1·2 ($P < 0·05$; one-tailed, single sample t test). The lack of co-operativity in cardiac TnC is consistent with the current view that it has only one regulatory Ca^{2+}-specific site. In skinned skeletal muscle n_H is usually higher than in skinned cardiac muscles (e.g. Godt & Nosek, 1989). Our results suggest that part of this difference may be explained by co-operativity in Ca^{2+} binding to skeletal TnC but not to cardiac TnC.

For both cardiac and skeletal muscle, the Ca^{2+} activation curves for the skinned fibres differed in two respects from the fluorescence curves for isolated TnC: the fibres had greater Ca^{2+} sensitivities (by 0·2–0·3 pCa units) and n_H values than found with the fluorescence curves of TnC. Both phenomena may be attributed to the influences of other proteins on the Ca^{2+} affinity of TnC *in situ* in the myofibrils. These factors include the following. (i) Interactions in the thin filament. It is known that the interaction of TnC with troponin I (TnI) enhances the Ca^{2+} affinity of the regulatory Ca^{2+} site up to 10-fold (Potter & Gergely, 1975; Holroyde *et al.* 1980; Grabarek *et al.* 1983), although on incorporation of TnI–TnC into the thin filament there may be a compensatory decrease in Ca^{2+} affinity (Zot *et al.* 1983). In addition, synergistic interactions between TnC molecules along the thin filament may explain the observed co-operativity of Ca^{2+} binding to TnC *in situ* (Grabarek *et al.* 1983) and of force development (Brandt, Diamond & Schachat, 1984; Güth & Potter, 1987). (ii) Interactions with myosin cross-bridges. The formation of strongly attached cross-bridges, either as rigor cross-bridges (Bremel & Weber, 1972), or as cycling cross-bridges during Ca^{2+} activation (Grabarek *et al.* 1983; Güth & Potter, 1987), increases the Ca^{2+} affinity of TnC in the intact filament. This can explain not only the higher Ca^{2+} affinity of TnC in the filaments compared with isolated TnC, but also the co-operativity in Ca^{2+} activation, since increasing Ca^{2+} activation of the thin filament would lead to the formation of more force-generating cross-bridges, thereby causing the Ca^{2+} affinity to rise. (iii) Non-equilibrium considerations. In addition to the above factors, which are apparent under equilibrium conditions (e.g. in binding studies), the non-equilibrium nature of the cross-bridge cycle can lead to n_H values that exceed the number of interacting sites (Shiner & Solaro, 1984). Similarly, the kinetics of the cross-bridge cycle can yield a higher Ca^{2+} sensitivity of force compared with Ca^{2+} binding, even if the Ca^{2+} affinity of TnC is unaffected; this is shown in the simple cross-bridge model given in the Appendix.

The role of troponin C in the modulation of calcium sensitivity

Potentially, alterations in the Ca^{2+} sensitivity of skinned fibres by factors such as pH, P_i, or caffeine could be brought about by alterations in the Ca^{2+} affinity of TnC. This study represents the first time, to our knowledge, that the properties of skinned muscles and the Ca^{2+} binding to the regulatory sites of isolated TnC have been examined under similar conditions of solution and

temperature; this makes it possible to suggest to what extent the changes in pCa_{50} in skinned fibres can be accounted for by direct effects on the Ca^{2+} affinity of TnC (Fig. 6). Of course, the Ca^{2+} sensitivities of fluorescence and force under control conditions have been studied previously after reincorporation of TnC_{DANZ} into skinned skeletal muscle myofilaments (e.g. Zot *et al.* 1986; Güth & Potter, 1987). We did not use this technique to investigate the effects of the above interventions, because (i) we were unable to reincorporate TnC_{IAANS} into skinned cardiac fibres (unpublished results) and knew of no report where this had been accomplished, and (ii) the results would have been very difficult to interpret, since a change in the Ca^{2+} sensitivity of TnC fluorescence could occur through several possible mechanisms: a direct effect on the Ca^{2+} affinity of TnC, indirect effects via thin filament protein interactions, or indirect effects via changes in cross-bridge attachment. A further problem is that fluorescence is enhanced to varying degrees, even in the absence of Ca^{2+}, depending upon the state of the attached cross-bridges: weakly bound; strongly bound, actively cycling; or strongly bound rigor cross-bridge (Güth & Potter, 1987; Morano & Rüegg, 1991). Thus, with intact myofilaments it would have been difficult to elucidate the contribution of direct effects on TnC to the modulation of the Ca^{2+} sensitivity of force production.

One potential problem with our study is that the cardiac TnC was isolated from a different species (cow) than used for the skinned muscles (rat). However, cardiac TnC is a highly conserved protein (Kobayashi, Takagi, Konishi, Morimoto & Ohsuki, 1989) and its properties would therefore be expected to be similar in the different mammalian species.

Skinned fibres showed a lowered Ca^{2+} sensitivity when the pH was reduced from 7·0 to 6·2, with falls in pCa_{50} of 1·09 and 0·55 in cardiac and skeletal fibres, respectively (Fig. 6). Under similar conditions, the pCa_{50} of isolated TnC_{IAANS} and TnC_{DANZ} decreased by 0·58 and 0·28. We derive two conclusions from these data. First, the direct effects of acidosis on TnC can account for some (about one-half) of the decrease in Ca^{2+} sensitivity in both cardiac and skeletal skinned fibres. Second, the 2-fold greater effect of pH on cardiac compared with skeletal fibres may be attributed to the 2-fold greater sensitivity of the cardiac TnC to pH changes (Fig. 6). In the case of acidosis, then, the properties of TnC clearly have a major part to play in the effects of acidosis on the Ca^{2+} activation of cardiac and skeletal myofibrils.

Since the direct action of protons on TnC could only account for one-half of the observed fall in the Ca^{2+} sensitivity of skinned fibres during acidosis (Fig. 6), the remainder must be due to indirect mechanisms. There are a number of these. First, there could be a fall in the TnI-induced enhancement of Ca^{2+} binding to TnC, as has been observed in skeletal preparations (El-Saleh & Solaro, 1988). In support of this role for TnI, the effect of pH on Ca^{2+} sensitivity increases from neonate to adult in the rat (Solaro, Lee, Kentish & Allen, 1988), when there are also changes in the isoforms of troponin I (TnI) and troponin T (TnT), but not of TnC. Second, acidosis may decrease the number of strongly bound, force-generating cross-bridges, and this would lower the Ca^{2+} affinity of TnC in the myofilament (see above). Third, some decrease in Ca^{2+} sensitivity could come about if H^+ affected certain reaction rates within the cross-bridge cycle in a similar manner to P_i (see Appendix). Such interactions within the myofilament that result in enhanced pH sensitivity can explain the difference between our results and those of Blanchard & Solaro (1984), who found that the total binding of Ca^{2+} to TnC in skinned cardiac fibres was sufficiently pH sensitive to account for *all* the effect of pH on the ATPase activity of intact myofibrils. Therefore, the acidosis-induced decrease in the Ca^{2+} affinity of isolated TnC must be enhanced by protein–protein interactions within the myofilament lattice.

The precise mechanism by which acidosis decreases the Ca^{2+} affinity of isolated TnC probably reflects a H^+-induced change in TnC structure similar to that induced by divalent metals (Lehrer & Leavis, 1974; Ingraham & Swenson, 1983). Thus, protons may compete directly with Ca^{2+} for binding at the low-affinity sites of TnC (possibly by increasing the Ca^{2+} 'off rate' from TnC). The 2-fold greater pH sensitivity of Ca^{2+} binding to cardiac TnC compared with skeletal TnC could also reflect structural differences between the two forms of TnC. Cardiac TnC has only the single regulatory Ca^{2+} binding site (site II), so it would seem probable that acidosis decreases the Ca^{2+} affinity of this site. We suggest that in skeletal TnC too, acidosis may affect the Ca^{2+} affinity of this site, leaving the other Ca^{2+}-specific site (site I) unaffected; then the overall pH-induced change in pCa_{50} produced by acidosis would be expected to be half of that in cardiac TnC, as observed (Fig. 6). An effect on site II alone could also explain the overall acidosis-induced decrease of n_H in skeletal TnC but not in cardiac TnC (Table 2). Thus, it seems possible that acidosis reduces the Ca^{2+} affinity of site II alone in both cardiac and skeletal TnC.

The decrease in Ca^{2+} sensitivity induced by P_i in cardiac and skeletal skinned fibres was not paralleled by any effect of P_i on Ca^{2+} binding to isolated cardiac and skeletal TnC (Fig. 6). The lack of effect on the Ca^{2+} affinity of TnC contrasts with a preliminary report by El-Saleh (1989), who found, using fluorescently labelled isolated TnC, that P_i desensitized both skeletal and cardiac TnC to Ca^{2+}. However, our results concur with preliminary data from Takayasu & Solaro (1990) using $^{45}Ca^{2+}$ binding measurements in skinned cardiac fibres. Our work indicates that the decrease in Ca^{2+} sensitivity of skinned fibres caused by P_i is not likely to be mediated by a direct reduction in the Ca^{2+} affinity of TnC, but needs the additional influences provided by other myofilament proteins.

It seems probable that a substantial part of the action of P_i on Ca^{2+} sensitivity is related to its effect of depressing maximum force production by reducing the number of force-generating cross-bridges, since this would indirectly lower the Ca^{2+} affinity of TnC in the myofilament (discussed above). An additional possibility is that there is a functional antagonism between Ca^{2+} and P_i in the cross-bridge cycle. There is good evidence that the Ca^{2+}-regulated step in the cross-bridge cycle is the P_i-release step (Chalovich, Chock & Eisenberg, 1981), i.e. the transition from weakly to strongly attached states. Consequently, Ca^{2+}–P_i antagonism would arise because an increase in $[Ca^{2+}]$ would promote the weak–strong transition, whereas an increase in $[P_i]$ would promote the reverse, strong–weak, transition. A model (Appendix) has been derived to show that, even if the Ca^{2+} affinity of TnC within the myofilament were constant, this phenomenon would produce a decrease in the Ca^{2+} sensitivity of skinned fibres. Thus, part of the decrease in Ca^{2+} sensitivity may be ascribed to a functional Ca^{2+}–P_i antagonism at the force-generating step in the cross-bridge cycle (see Appendix). However, both of the above mechanisms for the decrease in Ca^{2+} sensitivity predict that force and Ca^{2+} sensitivity should be reduced in parallel, and so cannot account for the findings that (i) in this study, P_i-induced depression of force was greater in cardiac muscle but the pCa_{50} was affected more in skeletal muscle, and (ii) the effect of P_i on Ca^{2+} sensitivity increases during development in rat cardiac muscle, whereas the effect on maximum force remains the same (Palmer & Kentish, 1992). This suggests that differences in myofibrillar proteins between cardiac and skeletal muscle, or isoform transitions during cardiac development, can influence the effect of P_i on Ca^{2+} sensitivity. An obvious possibility is that these variations are due to differences in other troponin components such as TnI.

Caffeine had no effect on the Ca^{2+} affinity of either cardiac or skeletal isolated TnC. Thus, a direct action of caffeine on TnC cannot explain the Ca^{2+} sensitization of skinned fibres, nor the 3-fold greater effect on Ca^{2+} sensitivity in cardiac muscle compared with skeletal muscle (Fig. 6). The action of caffeine on Ca^{2+} sensitivity is unlikely to be related to an effect on cross-bridge attachment or the weak–strong transition (cf. with pH or P_i), because (i) caffeine suppressed maximum force but increased Ca^{2+} sensitivity, and (ii) caffeine induced similar suppressions of maximum force in skeletal and cardiac preparations but had a much greater Ca^{2+}-sensitizing action in the latter. Supporting evidence that the action of caffeine is via a mechanism fundamentally different from that of pH or P_i, is that the effect of caffeine on Ca^{2+} sensitivity does not vary throughout development, whereas the sensitivity to pH or P_i increases (Palmer & Kentish, 1992). It may be that caffeine acts on troponin or tropomyosin within the intact thin filament to increase the Ca^{2+} affinity of TnC. An indirect effect to increase Ca^{2+} sensitivity by promoting cross-bridge attachment, for example, cannot be excluded, although it would be necessary to postulate a separate inhibitory action on the force produced by each cross-bridge in order to account for the observed reduction in maximum force production.

In conclusion, we have shown that the effects of acidosis, P_i and caffeine on Ca^{2+} sensitivity of skinned fibres vary between fast skeletal and cardiac muscle. The different Ca^{2+} sensitivities of the skinned fibres, and the differential actions of acidosis on Ca^{2+} sensitivity in cardiac and fast skeletal fibres, are attributable in part to the Ca^{2+} and H^+ sensitivities of cardiac and skeletal TnC. In contrast, the actions of P_i and caffeine cannot be explained by direct effects on Ca^{2+} binding to TnC, although the effects of P_i may be largely accounted for by its reversal of the P_i release step in the cross-bridge cycle.

APPENDIX

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A model of the interaction of P_i and Ca^{2+} on reaction rates in the cross-bridge cycle

The lack of effect of P_i on the Ca^{2+} affinity of isolated TnC indicates that the decrease in Ca^{2+} sensitivity of force is due to interactions between P_i and Ca^{2+} in the intact myofibril. Some of this is likely to be due to a decrease in the number of strongly attached force-generating cross-bridges, which are known to enhance the Ca^{2+} affinity of TnC (see Discussion). Such effects on the Ca^{2+} affinity of TnC, via protein–protein interactions, can be measured under equilibrium conditions (i.e. in binding studies). However, the cross-bridge cycle is a non-equilibrium, steady-state system, and non-equilibrium phenomena can lead to additional interactions. For example, a functional antagonism between Ca^{2+} and P_i on the *reaction rates* in the cross-bridge cycle would produce a possible additional mechanism by which P_i decreases the Ca^{2+} sensitivity of skinned fibres. This concept is illustrated here, using a very simple model of the cross-bridge cycle.

The model, presented for data relating to the skinned trabeculae, is shown in Fig. 7*A*, and the results from it are given in Fig. 7*B–D*. The model reflects the current understanding that the cross-bridge cycle has three types of cross-bridge state: detached (D); weakly attached, generating no force (W); and strongly attached, generating force (S). The transition between weakly and strongly attached is thought to be the reversible P_i release step. The model also incorporates the concept that Ca^{2+} regulation of the cycle occurs at the weak–strong transition (Chalovich *et al.* 1981). Although more recent evidence (e.g. Walker, Lu & Moss, 1992) suggests that a Ca^{2+}-regulated step immediately *preceding* the P_i release step is needed to explain kinetic data, in the steady-state situation considered here these two steps would be

functionally equivalent to a single step, as shown. From the scheme in Fig. 7 it can be seen that Ca^{2+} and P_i would have antagonistic effects, in that a rise in $[Ca^{2+}]$ will promote the W–S transition, whereas a rise in $[P_i]$ will have the reverse effect. It is this antagonism that, in the model, produces decreases in both force and Ca^{2+} sensitivity of the skinned fibres in the presence of P_i.

For the model, the following equations hold:

$$dW/dt = k_1D + k_{-2}[P_i]S - k_2W,$$

$$dS/dt = k_2W - k_{-2}[P_i]S - k_3S,$$

where rate constants k_1–k_3 are as shown in Fig. 7*A*.

$$dD/dt = k_3S - k_1D,$$

$$W + S + D = 1.$$

In the steady state, $dW/dt = dS/dt = dD/dt = 0$. From this, the relative concentration of the force-generating state, S, is given by: $[S] = 1/(k_3/k_1 + 1 + (k_3 + k_{-2}[P_i])/k_2)$.

The values chosen for the parameters (rate constants and K_d) were subject to the following constraints. (i) The rate of cross-bridge cycling at maximum Ca^{2+} activation was made equal to the ATPase activity of rat trabeculae measured under similar conditions (3·3 s^{-1}: Kentish & Stienen, 1994). (ii) At full Ca^{2+} activation the proportion of cross-bridges in state S was taken to be 0·75. (iii) The effect of 20 mM added P_i on maximum force, i.e. on [S], should equal the experimental data for cardiac muscle (Fig. 1). It was assumed that the W–S transition was the Ca^{2+}-regulated step, so its rate constant (k_2) was made a sigmoidal function of $[Ca^{2+}]$ according to the Hill equation (Fig. 7). To reproduce the slope of the experimental Ca^{2+}-activation curves, it was necessary to give n_H a value of 2·72 (as in Fig. 1). This implied co-operativity in the system, but it was assumed that this could arise without affecting the Ca^{2+} affinity of TnC (it may do so, however). This type of co-operativity could arise from nearest-neighbour interaction along the thin filament; the model does not incorporate any co-operativity due to effects of cross-bridge attachment on the Ca^{2+} affinity of TnC.

Values for the rate constants were chosen as follows. The maximum cycling rate of 3·3 s^{-1} is given by $k_3[S]$, so with $[S] = 0·75$ at full Ca^{2+} activation, k_3 must be 4·4 s^{-1}. The maximum value for k_2, termed $k_{2(max)}$, must be about 4 times greater than this to give a maximum [S] of 0·75, so $k_{2(max)}$ should be about 20 s^{-1}. The Ca^{2+} sensitivity of the system is determined by $k_{2(max)}$ and by the Ca^{2+} dissociation constant of TnC, K_d. With $k_{2(max)}$ of 20 s^{-1}, it was found that K_d had to be $\sim 10^{-5·3}$ M to fit the

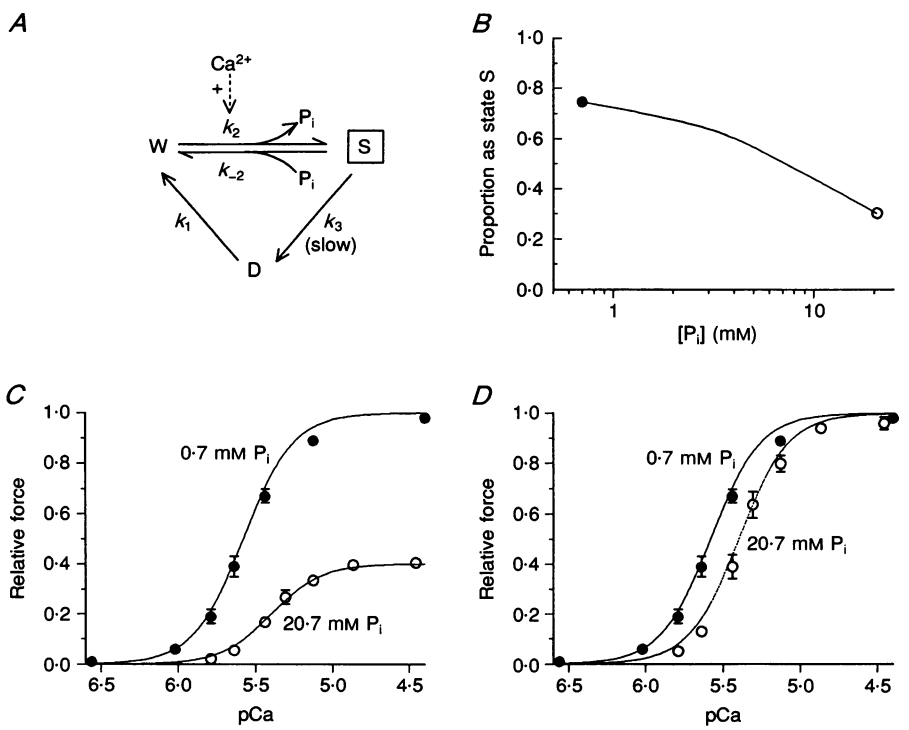


Figure 7. A model of the cross-bridge cycle to illustrate possible interaction between Ca^{2+} and P_i

The model scheme is shown in *A*, and its results in *B–D*. The model has three cross-bridge states: D, detached; W, weakly attached; and S, strongly attached. Only state S can generate force. The rate constants are: k_1, 100 s^{-1}; k_2, $20 \times [Ca^{2+}]^{n_H}/(K_d{}^{n_H} + [Ca^{2+}]^{n_H})$ s^{-1}; k_{-2}, $2000 \times [P_i]$ M^{-1} s^{-1}; k_3, 4·4 s^{-1}. K_d is $10^{-5·34}$ M and n_H is 2·72. The data points are those found experimentally with cardiac skinned fibres (Fig. 1) for: ●, no added P_i, calculated $[P_i]$ inside muscle = 0·7 mM (cf. Kentish, 1986); ○, 20 mM added P_i, calculated muscle $[P_i] = 20·7$ mM. The lines in *C* and *D* are derived from the model for 0·7 mM P_i and 20·7 mM P_i.

experimental pCa_{50} with zero added P_i. In fact, this value for K_d was very close to the measured value ($10^{-5.34}$ M) for isolated cardiac TnC_{DANZ} in zero P_i, so the latter value was chosen for the model. In the model P_i did not affect the Ca^{2+} affinity of TnC. The value for k_1 was arbitrarily set at 100 s^{-1}, since this step is thought to be fast. Its value does not influence the results. In fact, the detached state and k_1 could be omitted completely, but it was retained because cross-bridge detachment does occur during the cycle under most conditions. Finally, k_{-2} was varied so that the effect on maximum force of increasing $[P_i]$ from 0·7 mM (control) to 20·7 mM (20 mM added P_i) fitted the observed results (Fig. 1). This was the case when k_{-2} was $2000[P_i]$ s^{-1} (where $[P_i]$ is in molar terms).

From Fig. 7, it can be seen that the model was able to fit the experimental data well, and reproduced both the falls of maximum force and Ca^{2+} sensitivity induced by P_i. Note that there was no way that the effect of P_i on pCa_{50} could be manipulated independently of its effect on maximum force; rather, both were determined by k_{-2}. It may also be noted that the pCa_{50} for force need not be the same as the pCa_{50} for Ca^{2+} binding to TnC, which was ~0·3 pCa units lower than the resulting pCa_{50} of force.

The computed fall in pCa_{50} in the cardiac muscles was 0·17 in 20 mM P_i, compared with an actual shift of 0·19. This close similarity is likely to be fortuitous, and should not be taken as evidence that the interaction of Ca^{2+} and P_i in the model can account for more than a part of the depression of Ca^{2+} sensitivity. Indeed, fitting the model to other data (results not shown) illustrates that the Ca^{2+}–P_i antagonism could account for only (i) 73% of the 0·30 fall in pCa_{50} with skeletal fibres, and (ii) for only 50% of the 0·38 fall in pCa_{50} of cardiac fibres under different conditions (Kentish, 1986). Undoubtedly P_i also acts via other mechanisms, including the effect of strongly attached cross-bridges on the Ca^{2+} affinity of TnC, and the interactions with other myofilament proteins (see Discussion). Nevertheless, the model does illustrate that the depression of maximum force by P_i and a substantial part (perhaps one-half, or more) of the fall of Ca^{2+} sensitivity could arise through a common mechanism: a functional antagonism between the effects of Ca^{2+} and P_i on the kinetics of the force-generating step of the cross-bridge cycle. A similar mechanism could account for some of the acidosis-induced fall of maximum force and Ca^{2+} sensitivity, but as yet it is only speculation that the H^+ release step occurs close to the P_i release step in the cross-bridge cycle (Kentish, 1991).

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Note added in proof

Our conclusion, that the isoform of TnC determines the pH dependence of Ca^{2+} sensitivity, agrees with a recent study by J. M. Metzger, M. S. Parmacek, E. Barr, K. Pasyk, W.-I. Lin, K. L. Cochrane, L. J. Field & J. M. Leiden (*Proc. Nat. Acad. Sci. USA* **90**, 9036–9040 (1993), who studied the actions of acidosis on cardiac muscle that expressed sketal TnC.

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